

Defective Analysis & the Numbers

A Generator-Presentation Derivation of $\mathbb{N}, \mathbb{Z}, \mathbb{Q}, \mathbb{R}$, and $\mathbb{C}$

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Claim boundary. This paper uses ordinary mathematical language: generator, action, quotient, kernel, group completion, localization, valuation, metric completion, complexification, universal property, and presentation choice. The symbols $\theta, \alpha, \lambda, \rho$ are local notation for standard constructions and obstructions. The terminal claim about $\mathbb{C}$ is stated as a postulate with explicit proof obligations, not as a completed theorem.

Abstract

This note derives the classical scalar ladder

$$0 \longrightarrow \{0, 1\} \longrightarrow \mathbb{N} \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{R} \longrightarrow \mathbb{C}$$

from a minimal generating presentation. The natural numbers arise from the forward successor action. The integers arise by group completion. The rationals arise by localization at nonzero integers. The reals arise by completing the rational metric while correcting a presentation instability: rational closeness does not control dyadic generating load. The complex numbers arise by adjoining a polar unit so that exponentiation admits magnitude-preserving rotation rather than only real scaling. The final postulate is that, under the stated classical hypotheses, $\mathbb{C}$ is the terminal scalar layer obtained by adjoining additive inverses, multiplicative inverses, Archimedean completion, and polar complexification.

1. Notation and public-language replacement

Let

$$S_{\min} = \{0, 1\}$$

be the minimal generating set. The symbol 0 denotes the additive identity; 1 denotes the multiplicative identity and the additive generator.

This paper uses four symbols as compact labels:

θ = the additive successor/index action,

α = the multiplicative action generated by repeated addition,

λ = the completion obstruction exposed by rational closeness versus generating-presentation load,

ρ = the polarization obstruction exposed by real exponentiation.

These labels are not new mathematical structures. They mark standard transitions in the scalar ladder:

$$\boxed{\mathbb{N} = (\theta^+, \alpha^+), \quad \mathbb{Z} = (\theta^\pm, \alpha^+), \quad \mathbb{Q} = (\theta^\pm, \alpha^\pm), \quad \mathbb{R} = (\theta^\pm, \alpha^\pm, \lambda), \quad \mathbb{C} = (\theta^\pm, \alpha^\pm, \lambda, \rho).}$$

Here $+$ means only the forward operation is available; $\pm$ means the corresponding inverse operation has been adjoined.

Guiding principle. Each new number system is obtained by making a previously partial operation total, or by passing to a quotient/completion that removes presentation-dependence while preserving the intended scalar operations.

2. The zero stage and the minimal generating set

The construction begins with the distinction between no additive content and a unit generator:

$$S_{\min} = \{0, 1\}.$$

This is the smallest presentation from which the ordinary counting structure can be generated. The element 0 is neutral for addition; the element 1 generates the successor chain.

The set

$$S_2 = \{0, 1, 2\}$$

is not taken as primitive. It is the first successor-closed extension of $S_{\min}$ because

$$2 = 1 + 1.$$

This distinction will matter at the real-number stage: treating 2 as generated rather than primitive creates a measurable difference in dyadic presentation load.

3. The natural numbers: forward additive generation

Definition 1 (Natural-number presentation). *The natural numbers are generated from $S_{\min}$ by the forward successor action*

$$\theta(n) = n + 1.$$

Equivalently,

$$\mathbb{N} = \{0, 1, \theta(1), \theta^2(1), \dots\} = \{0, 1, 2, 3, \dots\}.$$

The additive structure is the free commutative monoid on one generator:

$$(\mathbb{N}, +, 0) = \langle 1 \rangle_{\text{cmon}}.$$

Multiplication is induced by repeated addition:

$$mn = \underbrace{n + n + \dots + n}_{m \text{ times}}.$$

Thus

$$(\mathbb{N}, +, \cdot, 0, 1)$$

is a commutative semiring.

The obstruction visible at this stage is that addition is not reversible. For $n > 0$, the equation

$$n + x = 0$$

has no solution in $\mathbb{N}$.

Hence the next scalar system is forced if additive cancellation is to become internal.

4. The integers: group completion

Definition 2 (Grothendieck group completion of $\mathbb{N}$). *Define*

$$\mathbb{Z} = \mathbb{N} \times \mathbb{N} / \sim$$

where

$$(a, b) \sim (c, d) \iff a + d = b + c.$$

The class of (a, b) is interpreted as the formal difference

$$[(a, b)] = a - b.$$

Then

$$n = [(n, 0)], \quad -n = [(0, n)].$$

Addition and multiplication are defined by

$$[(a, b)] + [(c, d)] = [(a + c, b + d)]$$

and

$$[(a, b)] \cdot [(c, d)] = [(ac + bd, ad + bc)].$$

These operations are well-defined on equivalence classes.

The result is the commutative ring

$$(\mathbb{Z}, +, \cdot, 0, 1).$$

The positive successor action has become an additive group action:

$$\theta^+ \rightsquigarrow \theta^\pm.$$

In the compact notation,

$$\boxed{\mathbb{Z} = (\theta^\pm, \alpha^+)}.$$

The obstruction visible at this stage is that multiplication is not reversible for non-units. For example,

$$2x = 1$$

has no solution in $\mathbb{Z}$.

Thus the next scalar system is forced if every nonzero scalar is to have a multiplicative inverse.

5. The rationals: localization by nonzero integers

Definition 3 (Localization of $\mathbb{Z}$). *Let*

$$S = \mathbb{Z} \setminus \{0\}.$$

The rational numbers are the localization

$$\mathbb{Q} = S^{-1}\mathbb{Z}.$$

Equivalently,

$$\mathbb{Q} = \mathbb{Z} \times (\mathbb{Z} \setminus \{0\}) / \sim$$

where

$$(a, b) \sim (c, d) \iff ad = bc.$$

The class of (a, b) is written

$$\frac{a}{b}.$$

The result is the field

$$(\mathbb{Q}, +, \cdot, 0, 1).$$

Every nonzero element is invertible:

$$q \neq 0 \implies q^{-1} \in \mathbb{Q}.$$

The multiplicative action has become reversible:

$$\alpha^+ \rightsquigarrow \alpha^\pm.$$

Thus

$$\boxed{\mathbb{Q} = (\theta^\pm, \alpha^\pm)}.$$

At this stage, the ordinary algebraic inverse problem has been solved. The next obstruction is not a missing inverse. It is a mismatch between rational metric closeness and presentation load.

6. The completion obstruction: rational closeness versus presentation load

The rational numbers carry the usual absolute-value metric

$$d(x, y) = |x - y|.$$

This metric detects numerical closeness. It does not, by itself, control the complexity of how a rational number is generated from a chosen presentation.

The relevant presentation issue is already visible in the treatment of 2.

There are two natural generating presentations:

$$P_{\min} = \{0, 1\},$$

where 2 is generated as $1 + 1$, and

$$P_2 = \{0, 1, 2\},$$

where 2 is treated as already available after the first successor extension.

For $x \in \mathbb{Q}^\times$, write its dyadic decomposition as

$$x = 2^{\nu_2(x)} \frac{a}{b},$$

where a, b are odd integers and $\nu_2(x) \in \mathbb{Z}$ is the dyadic valuation.

Define the dyadic generating load relative to the two presentations by

$$L_{P_{\min}}^{(2)}(x) = |\nu_2(x)|,$$

while, for the presentation in which powers of 2 are treated as primitive dyadic structure,

$$L_{P_2}^{(2)}(x) = 0.$$

The second load records only the dyadic component. It does not claim that all other factors of x are free.

Proposition 1 (Rational closeness does not bound dyadic generating load). *Let*

$$b_n = 2^{-n} \in \mathbb{Q}.$$

Then

$$|b_n - 0| = 2^{-n} \rightarrow 0,$$

but

$$L_{P_{\min}}^{(2)}(b_n) = n \rightarrow \infty,$$

while

$$L_{P_2}^{(2)}(b_n) = 0.$$

Proof. Since $b_n = 2^{-n}$, its dyadic valuation is

$$\nu_2(b_n) = -n.$$

Therefore

$$L_{P_{\min}}^{(2)}(b_n) = |\nu_2(b_n)| = n.$$

At the same time, the usual rational metric gives

$$|b_n| = 2^{-n} \rightarrow 0.$$

Under the presentation in which the dyadic component is primitive, the dyadic load is recorded as zero by definition. $\square$

The conclusion is:

Convergence in the rational metric does not control generating-presentation load.

This is the obstruction labelled λ in this note.

Definition 4 (λ -completion). *The symbol λ denotes the transition from $\mathbb{Q}$ to a scalar system in which rational Cauchy convergence has endpoints and presentation choices that vanish in the rational metric are identified at the scalar level.*

The standard public construction realizing this transition is the Cauchy completion.

Let

$$C(\mathbb{Q}) = \{(x_n)_{n \geq 0} : x_n \in \mathbb{Q} \text{ and } (x_n) \text{ is Cauchy}\}.$$

Let

$$c_0(\mathbb{Q}) = \{(x_n) \in C(\mathbb{Q}) : x_n \rightarrow 0\}.$$

Then define

$$\mathbb{R} = C(\mathbb{Q})/c_0(\mathbb{Q}).$$

Two rational Cauchy presentations determine the same real number exactly when their difference tends to zero:

$$(x_n) \sim (y_n) \iff x_n - y_n \rightarrow 0.$$

Thus a real number is a scalar equivalence class of rational approximations, not a single rational presentation.

In compact notation,

$$\mathbb{R} = (\theta^\pm, \alpha^\pm, \lambda).$$

7. The real numbers as the λ -repair

Operations descend to the quotient by null sequences:

$$[x] + [y] = [x + y],$$

$$[x][y] = [xy],$$

$$-[x] = [-x].$$

The embedding of rationals is

$$\iota : \mathbb{Q} \hookrightarrow \mathbb{R}, \quad q \mapsto [(q, q, q, \dots)].$$

The quotient is a field. If $[x] \neq 0$, then x is not a null sequence. Since x is Cauchy, it is eventually bounded away from zero: there are $\varepsilon > 0$ and N such that

$$n > N \implies |x_n| > \varepsilon.$$

Define a rational Cauchy sequence y by

$$y_n = \begin{cases} 1, & n \leq N, \\ 1/x_n, & n > N. \end{cases}$$

Then

$$xy - 1 \in c_0(\mathbb{Q}),$$

so

$$[x][y] = [1].$$

Hence every nonzero real class has an inverse.

The ordered-field structure is obtained by declaring

$$[x] > 0$$

when some representative is eventually bounded below by a positive rational:

$$\exists \varepsilon \in \mathbb{Q}_{>0} \exists N \forall n > N, \quad x_n > \varepsilon.$$

The construction yields the complete Archimedean ordered field.

Interpretive point. The move from $\mathbb{Q}$ to $\mathbb{R}$ is not merely the assertion that irrational numbers exist. It is the construction of a scalar system where rational metric convergence is allowed to have stable endpoints while presentation-dependent loads, such as the dyadic load of 2^{-n} , no longer determine scalar equality.

8. Exponentiation as a coupled additive-multiplicative action

After $\mathbb{R}$ is obtained, both addition and multiplication are reversible in the usual scalar sense, and metric completion is available. Yet a remaining obstruction appears when additive parameters are coupled to multiplicative action by exponentiation.

Real exponentiation gives

$$\exp : \mathbb{R} \rightarrow \mathbb{R}_{>0}, \quad \exp(t) = e^t.$$

This is a one-parameter multiplicative subgroup of positive real scalars:

$$e^{s+t} = e^s e^t.$$

However, it is line-bound:

$$e^t > 0 \quad \text{for all } t \in \mathbb{R}.$$

It scales. It does not rotate.

If the scalar system is required to support a magnitude-preserving polar one-parameter action, the real line is insufficient. One needs an element whose exponential generates rotation rather than positive scaling.

This is the obstruction labelled ρ in this note.

9. The complex numbers: polar complexification

Definition 5 (Complexification of $\mathbb{R}$). *Adjoin an element i satisfying*

$$i^2 = -1.$$

Equivalently,

$$\mathbb{C} = \mathbb{R}[x]/(x^2 + 1).$$

Every element has the form

$$z = a + bi, \quad a, b \in \mathbb{R}.$$

As a real vector space,

$$\mathbb{C} \cong \mathbb{R}^2.$$

Multiplication is

$$(a, b)(c, d) = (ac - bd, ad + bc).$$

The polar unit repairs the exponentiation obstruction:

$$e^{it} = \cos t + i \sin t.$$

This satisfies

$$|e^{it}| = 1,$$

and therefore gives a magnitude-preserving rotation action:

$$z \mapsto e^{it} z.$$

Thus the transition

$$\mathbb{R} \longrightarrow \mathbb{C}$$

is not only the algebraic solution of

$$x^2 + 1 = 0.$$

That equation is the minimal polynomial symptom. The structural repair is polar complexification: real scaling is extended to include unit-magnitude rotation.

In compact notation,

$$\boxed{\mathbb{C} = (\theta^\pm, \alpha^\pm, \lambda, \rho).}$$

The standard algebraic fact behind the terminal role of $\mathbb{C}$ is that $\mathbb{C}$ is algebraically closed: every nonconstant polynomial with complex coefficients has a complex root. This fact is used here only inside the claim boundary stated below.

10. The full scalar ladder

The complete construction is

$$\boxed{0 \longrightarrow S_{\min} \longrightarrow \mathbb{N} \longrightarrow \mathbb{Z} \longrightarrow \mathbb{Q} \longrightarrow \mathbb{R} \longrightarrow \mathbb{C}.}$$

Equivalently,

$$\boxed{0 \longrightarrow \{0, 1\} \longrightarrow \theta^+ \longrightarrow \theta^\pm \longrightarrow \alpha^\pm \longrightarrow \lambda \longrightarrow \rho.}$$

System	Available structure	Construction added	Public reading
0	no scalar distinction	none	null starting stage
$S_{\min}$	0, 1	minimal generating set	identity and unit generator
$\mathbb{N}$	θ^+, α^+	successor closure	forward additive generation; semiring
$\mathbb{Z}$	$\theta^\pm, \alpha^+$	group completion	additive inverses; ring
$\mathbb{Q}$	$\theta^\pm, \alpha^\pm$	localization	multiplicative inverses; field
$\mathbb{R}$	$\theta^\pm, \alpha^\pm, \lambda$	metric completion	stable endpoints of rational Cauchy approximation; complete ordered field
$\mathbb{C}$	$\theta^\pm, \alpha^\pm, \lambda, \rho$	complexification	polar unit; algebraically closed field

11. Universal properties

Each transition can be stated by a universal property.

$\mathbb{N}$

The additive monoid $(\mathbb{N}, +, 0)$ is the free commutative monoid on one generator. For every commutative monoid M and every element $m \in M$, there is a unique monoid homomorphism

$$f : \mathbb{N} \rightarrow M$$

with

$$f(1) = m.$$

$\mathbb{Z}$

The integers are the group completion of the additive monoid $\mathbb{N}$. For every abelian group G and monoid homomorphism

$$f : \mathbb{N} \rightarrow G,$$

there is a unique group homomorphism

$$\bar{f} : \mathbb{Z} \rightarrow G$$

such that f factors through $\mathbb{Z}$.

$\mathbb{Q}$

The rationals are the localization of $\mathbb{Z}$ at the nonzero integers. For every ring R and ring homomorphism

$$f : \mathbb{Z} \rightarrow R$$

that sends every nonzero integer to a unit of R , there is a unique ring homomorphism

$$\bar{f} : \mathbb{Q} \rightarrow R$$

through which f factors.

$\mathbb{R}$

The reals are the metric completion of $\mathbb{Q}$ under the usual absolute value. Every uniformly continuous map from $\mathbb{Q}$ into a complete metric space extends uniquely to $\mathbb{R}$ when the required hypotheses hold.

In this paper, the completion is motivated by the additional presentation fact that rational closeness does not control dyadic generating load.

$\mathbb{C}$

The complex numbers are the quadratic extension

$$\mathbb{R}[x]/(x^2 + 1)$$

and the algebraic closure of $\mathbb{R}$. They also supply the one-dimensional polar unitary group

$$U(1) = \{z \in \mathbb{C} : |z| = 1\} = \{e^{it} : t \in \mathbb{R}\}.$$

12. Terminal scalar postulate

Postulate 1 (Terminal classical scalar postulate). *Subject to the following hypotheses:*

1. *the system is generated from $S_{\min} = \{0, 1\}$;*
2. *addition and multiplication are commutative scalar operations with identities;*
3. *additive inverses are adjoined by group completion;*
4. *multiplicative inverses for nonzero scalars are adjoined by localization;*
5. *rational metric convergence is completed in the Archimedean ordered sense;*
6. *exponentiation is extended to include magnitude-preserving polar rotation;*
7. *equivalence is taken up to isomorphism or harmless change of presentation;*

then $\mathbb{C}$ is the terminal classical scalar layer forced by these requirements.

This postulate does not claim that no other algebraic systems exist. It does not exclude finite fields, p -adic fields, nonstandard real fields, surreal numbers, quaternions, Clifford algebras, matrix algebras, operator algebras, schemes, sheaves, or higher categories. Those systems either alter the hypotheses, change the notion of scalar, abandon commutativity, use a different absolute value, add dimension rather than scalar closure, or move into a different category.

The postulate claims only that the ladder

$$\mathbb{N} \rightarrow \mathbb{Z} \rightarrow \mathbb{Q} \rightarrow \mathbb{R} \rightarrow \mathbb{C}$$

is terminal for the stated classical scalar program.

13. Open questions and proof obligations

The terminal postulate is intentionally strong. A complete proof requires at least the following work.

1. **Category of presentations.** Define the category whose objects are scalar presentations and whose morphisms preserve 0, 1, addition, multiplication, completion data, and polar structure.
2. **Presentation-load invariance.** Prove which generating-load functions are invariantly meaningful, and which are artifacts of a chosen presentation.
3. **Prime and valuation generality.** The dyadic example uses ν_2 . One should determine whether analogous p -adic valuations ν_p produce distinct completion obstructions or whether the Archimedean completion absorbs them only under additional hypotheses.
4. **Equivalence of completions.** Relate the Cauchy construction, Dedekind cuts, nested intervals, and other constructions of $\mathbb{R}$ inside the presentation-load argument.
5. **Polarization criterion.** Specify the exact universal property that forces $\mathbb{C}$ from $\mathbb{R}$: algebraic closure, polar exponentiation, two-dimensional real rotation, or a combined condition.
6. **Boundary systems.** Classify precisely why p -adics, quaternions, nonstandard reals, surreal numbers, and operator algebras fall outside the terminal scalar postulate or require a modified postulate.
7. **Terminality theorem.** Convert the postulate into a theorem by proving that any scalar system satisfying the hypotheses factors through $\mathbb{C}$ uniquely up to the chosen equivalence.

14. Final compression

The derivation can be compressed to one line:

$$\mathbb{C} = S_{\min} + \theta^{\pm} + \alpha^{\pm} + \lambda + \rho.$$

Expanded in public language:

$\mathbb{C}$ = minimal generating set + additive inverses + multiplicative inverses + Archimedean completion + polar complexification.
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Thus the number systems are not presented as a list of unrelated enlargements. They are a sequence of forced repairs to partial scalar structure:

$\mathbb{N}$ generates; $\mathbb{Z}$ reverses addition; $\mathbb{Q}$ reverses multiplication; $\mathbb{R}$ completes rational approximation; $\mathbb{C}$ polarizes exponentiation.
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